

DATA PRODUCT CARD

NYC Taxi Monthly Borough Statistics

Curated Gold-layer data product for monthly NYC Yellow Taxi analysis by pickup and dropoff borough. The product is prepared for consumption by another student through DuckDB, CSV/Parquet export and a documented data contract.

Field	Description
Product owner	Wiktor Lenarczyk - MS Teams / PJATK
Product purpose	Make raw NYC Yellow Taxi data easy to consume for dashboards, revenue analysis and borough-to-borough mobility analysis.
Business problem solved	Raw taxi Parquet files are large and difficult to query directly. This product exposes an aggregated, documented and quality-checked Gold table.
Data sources	NYC TLC Yellow Taxi Trip Records; NYC TLC Taxi Zone Lookup CSV.
Pipeline	Python + DuckDB + SQL + Dagster; Bronze -> Silver -> Gold medallion architecture.
Refresh frequency	On demand after each pipeline run; can be scheduled daily/monthly.
Access method	DuckDB table warehouse.duckdb: gold_monthly_borough_stats; optional CSV/Parquet export in /export.

Schema - Gold table: gold_monthly_borough_stats

Column	Type	Description
trip_month	DATE/TIMESTAMP	Month of taxi pickup; aggregation key.
pickup_borough	VARCHAR	NYC borough where trip started.
dropoff_borough	VARCHAR	NYC borough where trip ended.
trip_count	BIGINT	Number of valid Silver trips.
avg_trip_distance	DOUBLE	Average trip distance in miles.
avg_total_amount	DOUBLE	Average total amount in USD.
total_revenue	DOUBLE	Total amount in USD for the group.

Example consumption

```
SELECT trip_month, pickup_borough, SUM(trip_count) AS trips, SUM(total_revenue) AS revenue
FROM gold_monthly_borough_stats
GROUP BY trip_month, pickup_borough
ORDER BY trip_month, trips DESC;
```

DATA QUALITY, LIMITATIONS AND PUBLICATION

Key quality metrics are calculated after every pipeline run by scripts/generate_data_product_metrics.py and saved to quality/data_quality_metrics.csv. Last successful local run: 2026-05-10 20:05 Europe/Warsaw. Result: 4 PASS metrics and 1 WARN freshness metric because the downloaded TLC files are historical.

Metric	Definition	Current value	Threshold	Cadence
Gold row count	Number of records in the Gold table.	3312 (PASS)	> 0	Every pipeline run
Gold key completeness	% rows where trip_month, pickup_borough and dropoff_borough are not null.	100.0% (PASS)	>= 95%	Every pipeline run
Positive trip count validity	% rows where trip_count is greater than 0.	100.0% (PASS)	100%	Every pipeline run
Non-negative amount validity	% rows where avg_total_amount and total_revenue are non-negative.	100.0% (PASS)	>= 99%	Every pipeline run
Data freshness months	Months between current date and latest trip_month.	74 months (WARN - historical source files)	<= 6 months if current files are downloaded	Every pipeline run

Known limitations

Data freshness is WARN in this run because the source files are historical rather than current operational data; this does not block historical analysis.

- The product depends on downloaded NYC TLC source files; repository users must run the pipeline or use exported CSV/Parquet files.
- Incremental mode tracks execution state and avoids repeated downloads, but Gold rebuild is still table-level rather than strict partition-only processing.
- Borough values depend on successful join with taxi_zone_lookup.csv; unknown or changed zone IDs may produce null boroughs.
- The product is monthly aggregated and is not intended for trip-level investigation.
- Fare outliers may remain if they pass basic validity rules.

MS Teams Data Marketplace publication checklist

- Upload this Data Product Card as DOCX and PDF.
- Add link to Git repository or data_product_contract.yaml.
- Add quality/data_quality_metrics.csv after pipeline run.
- Add dataset access instructions or export/gold_monthly_borough_stats.csv / .parquet.
- If the repository is private, state that access is available by invitation or through the exported dataset in Teams.

Short marketplace description: Monthly NYC Yellow Taxi demand and revenue statistics by pickup and dropoff borough. Useful for dashboarding, borough comparison, revenue analysis and demonstrating a medallion data product with quality metrics and a data contract.